

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 17

15 July 2025

Overview:

Today's session introduced us to the world of **Natural Language Processing (NLP)** — a branch of artificial intelligence that focuses on enabling computers to understand, interpret, and generate human language. We explored the importance of NLP in modern applications like chatbots, translation systems, and voice assistants.

Learning Objectives:

- Understand what NLP is and why it matters
- Learn about the basic steps in NLP workflows
- Explore key concepts like tokenization, stemming, and lemmatization
- Get an overview of real-world applications and tools used in NLP

What is NLP?

NLP stands for Natural Language Processing, which allows machines to read, understand, and derive meaning from human language. It combines linguistics, computer science, and AI to bridge the gap between human communication and computer understanding.

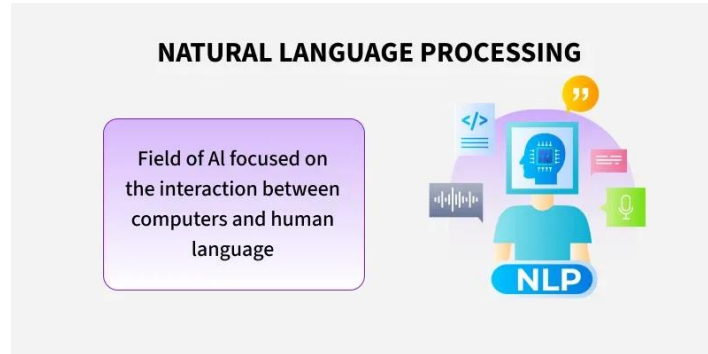


Fig. NLP

Importance of NLP:

Natural Language Processing plays a vital role in bridging the gap between human language and computer understanding. It helps machines interpret, analyze, and generate text or speech just like humans do. NLP is used in many real-life applications such as chatbots, voice assistants (like Siri and Alexa), language translation tools, spam filters, and sentiment analysis. By enabling machines to understand human language, NLP makes communication with technology more natural, efficient, and user-friendly.

NLP Techniques:

- Tokenization: Splitting text into individual words or tokens
- Stemming & Lemmatization: Reducing words to their root forms
- Stopword Removal: Ignoring common words like “is”, “the”, “and”
- Bag of Words / TF-IDF: Representing text in numerical form for models
- Named Entity Recognition (NER): Identifying names, locations, dates, etc.

Conclusion:

Today's session showed how powerful NLP is in making machines language aware. From virtual assistants to smart email replies, NLP is behind the scenes making machines interact more naturally with us. This was a very interesting topic that showed us how computers can understand our language.